

Question Number	Answer	Additional Guidance	Mark
1a	A description that includes six of the following points: • (compare) pollen grains with and without stigma extract • method of measuring growth of pollen tube (1) • identification of one variable with suitable method of control (1) • identification and method of control of a second variable (1) • stated time (intervals) for measuring length of pollen tube (1) • method for calculating rate of growth (1) • repeats (for each treatment) and calculate a mean (1)	Accept with extract and with sucrose as control Accept different concentrations of extract as long as one is zero Need to mention microscope Accept use of rule, squared paper, stage micrometer Accept {temperature / pH / volume of water / light intensity / age of plant / age of pollen grains / time after ripening of stamens, Ignore size of pollen grains Accept measure after a single suitable interval eg after 30 mins min, max 12 hours; or at suitable intervals eg 10-30 mins <u>Change in length</u> time Accept repeats and calculate SD	6

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1(b)	A description that includes four of the following: - starch is {digested / broken down / hydrolysed} to release glucose (1) - condensation reactions are used to form pectin (1) - pectin {secreted into / enclosed in / transported in} vesicles (1) - vesicles fuse with the {tip / cell membrane} to release {pectin / contents} (1) (respiration in) mitochondria releases {ATP / energy} for production of {vesicles / pectin / enzymes / other relevant molecules} (1)	Accept release monosaccharides Accept glucose joined by glycosidic bonds to form pectin/polysaccharide Accept reference to exocytosis releasing contents Accept energy used for movement of vesicles / exocytosis Piece together mp5	